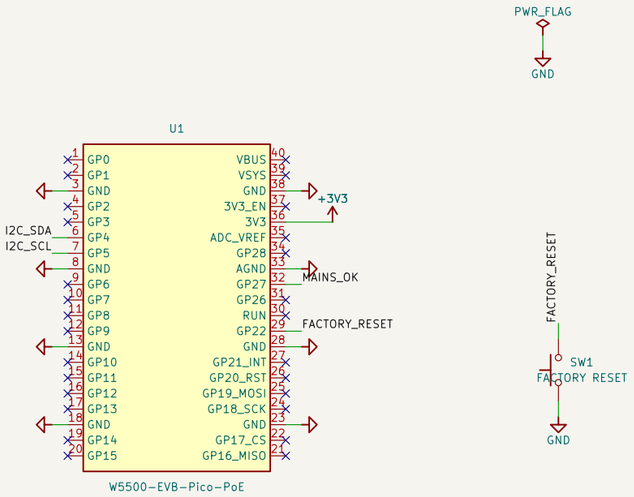
## W5500-EVB-Pico-PoE (plugs into 2x 1x20 female headers)

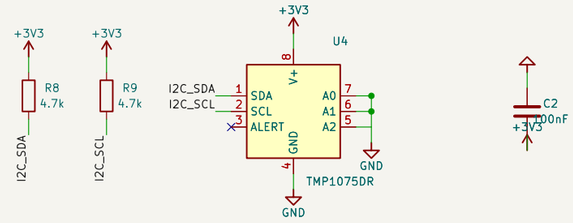






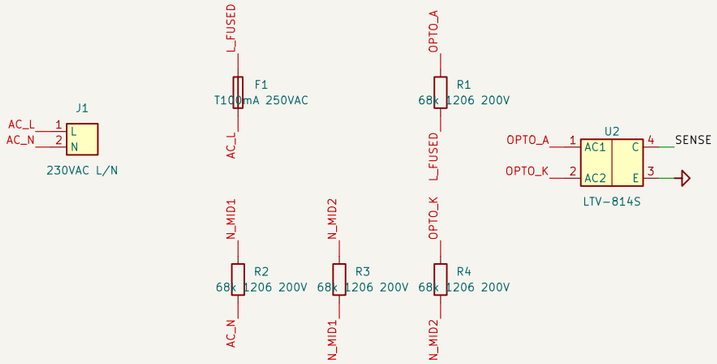
GP16..GP21 are used by the on-board W5500 - do not use.
GP27 (pin 32) = MAINS_OK input: HIGH = 230VAC present.
GP4/GP5 = I2C0 SDA/SCL to TMP1075 (address 0x48).
GP22 (pin 29) = SW1 to GND: hold 10 s for a factory reset (internal pull-up).
+3V3 comes from the module's 3V3(OUT) pin 36 (PoE powered).

## Temperature sensor TMP1075 (I2C 0x48, +-1 C)



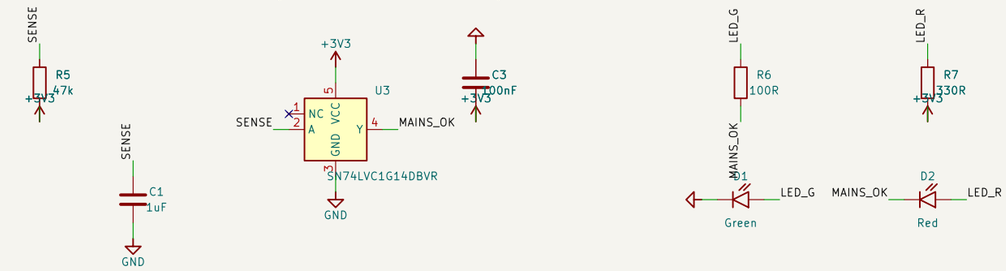

Placed away from the module and the mains resistors.
ALERT not used.

## HAZARDOUS VOLTAGE - 230VAC MAINS SENSE




F1 T100mA 250VAC in the only current loop. 4x 68k 1206 (>=200V): ~1.2 mA peak, ~50 mW each.
LTV-814S AC-input optocoupler, 5 kVrms. PCB: >=6.5 mm clearance, >=6.4 mm creepage + routed slot.

## Mains detect + LED indication (works without firmware)




Opto pulls SENSE low while mains is present; C1 holds it low over the zero crossings (47k x 1uF = 47 ms).
MAINS_OK = NOT SENSE: green LED on = mains ON, red LED on = mains OFF.




Mains detect on GP27, LED status, TMP1075 temperature, factory reset button on GP22
**Kenneth Foli Jørgensen**

Sheet: /
File: pico-poe-mains-carrier.kicad_sch

***Title: RF-UPS-Monitor carrier for the W5500-EVB-Pico-PoE***

| Size: A3 | Date: 2026 | **Rev: 1.1** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |